

0.1 Question 7 (4 points)

Areas under curves will be used in this class when we calculate some probabilities.

Consider the function $f(x)$ shown below:

Calculate $\int_{3.25}^6 f(x) dx$. You can use geometry and/or calculus. Show all steps/explain your reasoning.

If you use the equation of a line in your calculation, you must show all steps deriving the equation of that line to receive credit.

Question 7 Solution) Type your work answering to **Question 7** in this cell (use LaTeX for any math equations). Show all of your steps and fully justify your answer. Do not add any additional cells to this part.

```
In [40]: from scipy.special import comb
```

```
        comb(10, 3)
```

```
        # Optional cell for you to do calculations/check your work. All work must be written in LaTeX
```

```
Out[40]: 120.0
```


0.2 Question 8b (3 points)

What is the probability that if I roll two 6-sided dice they add up to **at most** 9? Use LaTeX (not code) in the cell directly below to show all of your steps and fully justify your answer.

Type your answer here, replacing this text.

0.3 Question 8c (3 points)

Suppose you uncharacteristically show up to a quiz completely unprepared. The quiz has 10 questions, each with 5 multiple choice options. You decide to guess each answer in a completely random way. What is the probability that you get exactly 3 questions correct? Use Markdown and LaTeX (not code) in the cell directly below to show all of your steps and fully justify your answer.

Type your answer here, replacing this text.

0.4 Question 8d (3 points)

A 9-bit string is sent over a network. The receiver only accepts strings that either start with 01 and/or end with 100. How many unique 9-bit strings will the receiver accept? Explain/justify using math.

Enter your answer for part 8d) in this cell (double click on this cell and write all steps using Markdown and LaTeX):

